Solve with us 3.2 - Not Graded

This assignment will not be graded and is only for practice.
Key points: 1Key points: 1

- Definition of linear dependence: A set of vectors $v_1, v_2, ...v_n \in V$v1, v2, ...vn ∈ V is said to be linearly dependent if there exist scalars $a_1, a_2, ...a_n \in R$a1, a2, ...an ∈ R, not all zero such that $a_1 v_1 + a_2 v_2 + ... + a_n v_n$ = 0a1v1 + a2v2 + ... + anvn = 0.
- To check the vectors $v_1, v_2, ...v_n \in R^m$ v1, v2, ...vn ∈ Rm (with usual addition and scalar multiplication) are linearly dependent, we have to verify that the homogeneous system of linear equations Ax = 0Ax = 0 has infinitely many solutions, where the jth jth column of AA is v_j vj.
- Definition of linear independence: A set of vectors $v_1, v_2, ...v_n \in V$ v1, v2, ...vn ∈ V is said to be linearly independent if $a_1 v_1 + a_2 v_2 + ... + a_n v_n$ = 0a1v1 + a2v2 + ... + anvn = 0 implies that a_i = 0 ai = 0 for i = 1, 2, ...ni = 1, 2, ...n.
- To check the vectors $v_1, v_2, ...v_n \in R^m$ v1, v2, ...vn ∈ Rm (with usual addition and scalar multiplication) are linearly independent, we have to verify that the homogeneous system of linear equations Ax = 0Ax = 0 has only the trivial solution (i.e., x = 0x = 0), where the jth jth column of AA is v_j vj.
- If SS is a set containing zero vector, then the set SS is a linearly dependent set.
- Let SS be a set of vectors from a vector space VV. If an element of SS is a scalar multiple of another element from the set SS, then SS is a linearly dependent set.

1 point

Vectors (2,1) and (6, 3) from R^2 R2 are

- ○ linearly independent.
- ○ linearly dependent.
- ○ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
linearly dependent.

1 point

Suppose the set $\{(3, 7, 4), v\}$ {(3, 7, 4), v} is a linearly dependent set in the vector space R^3 R3 with usual addition and scalar multiplication. Which of the following vectors are possible candidates for vv?

- ☐ (0, 0, 0)(0, 0, 0)
- ☐ $(5, \frac{35}{3}, \frac{20}{3})$(5, 335, 320)
- ☐ (1, 0, 0)(1, 0, 0)
- ☐ (6, 14, − 8)(6, 14, −8)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(0, 0, 0)(0, 0, 0)
$(5, \frac{35}{3}, \frac{20}{3})$(5, 335, 320)

1 point

The set of vectors $\{(1, 1, -1), (-1, 1, 1),$ and $(0, \frac{1}{2}, 0)\}$ {(1, 1, −1), (−1, 1, 1), and (0, 21, 0)} in ... is

- ○ linearly independent.

○ linearly dependent.
○ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
linearly dependent.

1 point

$M_{2\times 2}(R)$M2×2(R) denotes the vector space consisting of real square matrices of order 2 with usual matrix addition and scalar multiplication. Consider the following matrices:

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, M_3 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \text{M1} = [100{-}1], \text{M2} = [10{-}10], \text{M3} = [1{-}101]$$

Which of the following options is(are) true?

[Hint:Hint: Form a system of linear equations using matrix equation $x\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + y\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$x [a11a21 a12a22] + y [b11b21b12b22] = 0]

☐ $\{M_1, M_2\}$ {M1, M2} is a linearly independent set.
☐ $\{M_1, M_2, M_3\}$ {M1, M2, M3} is a linearly dependent set.
☐ $\{M_2, M_3\}$ {M2, M3} is a linearly dependent set.
☐ $\{M_1, M_3\}$ {M1, M3} is a linearly independent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{M_1, M_2\}$ {M1, M2} is a linearly independent set.
$\{M_1, M_3\}$ {M1, M3} is a linearly independent set.

Key points: 2Key points: 2

- Given two vectors (x_1, y_1), and $(x_2, y_2) \in R^2$ (x1, y1), and (x2, y2) ∈ R2 construct $A = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}$ A = [x1y1x2y2]. Then these vectors are linearly dependent if and only if $det(A) = 0$det(A) = 0 and linearly independent if and only if $det(A) \neq 0$det(A) = 0.
- Given three vectors $(x_1, y_1, z_1), (x_2, y_2, z_2)$ and $(x_3, y_3, z_3) \in R^3$ (x1, y1, z1), (x2, y2, z2) and (x3, y3, z3) ∈ R3 construct $A = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$A = [x1y1z1x2y2z2x3y3z3]. Then these vectors are linearly dependent if and only if $det(A) = 0$det(A) = 0 and linearly independent if and only if $det(A) \neq 0$det(A) = 0

1 point

Vectors (2,3) and (4, 5) from R^2 R2 are

[Hint:Hint: Form the matrix using the vectors and calculate the determinant of the matrix.]

○ linearly independent.
○ linearly dependent.
○ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
linearly independent.

1 point

Consider a set of vectors S = {(1, 2), (− 1, 2), (3, 1), (− 3, 1)} ⊂ S = {(1, 2), (−1, 2), (3, 1), (−3, 1)} ⊂ R^2 R2. Choose the set of correct options.

[Hint:Hint: Form the matrix using the vectors as the columns for each option and calculate the determinant of the matrix.]

☐ {(1, 2), (− 1, 2)}{(1, 2), (−1, 2)} is a linearly dependent set.

☐ {(− 1, 2), (− 3, 1)}{(−1, 2), (−3, 1)} is a linearly independent set.

☐ {(− 3, 1), (− 1, 2)}{(−3, 1), (−1, 2)} is a linearly independent set.

☐ {(3, 1), (− 3, 1)}{(3, 1), (−3, 1)} is a linearly dependent set.

☐ {(3, 1), (1, 2)}{(3, 1), (1, 2)} is a linearly independent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
{(− 1, 2), (− 3, 1)}{(−1, 2), (−3, 1)} is a linearly independent set.
{(− 3, 1), (− 1, 2)}{(−3, 1), (−1, 2)} is a linearly independent set.
{(3, 1), (1, 2)}{(3, 1), (1, 2)} is a linearly independent set.

1 point

The set of vectors {(1, 1, 0), (0, 1, 1), (1, 0, 1)}{(1, 1, 0), (0, 1, 1), (1, 0, 1)} is

[Hint:Hint: Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

○ linearly dependent.
○ linearly independent.
○ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
linearly independent.

1 point

Suppose {(2, 3, 0), (0, 1, 0), v}{(2, 3, 0), (0, 1, 0), v} is a linearly dependent set in the vector space R^3 R3 with usual addition and scalar multiplication. Which of the following vectors are possible candidates for vv?

☐ (2, 3, 1)(2, 3, 1)

☐ (1, 2, 0)(1, 2, 0)

☐ (1, 3, 0)(1, 3, 0)

☐
$(0, 1, 1)$(0, 1, 1)
☐
$(\pi, e, 0)$(π, e, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(1, 2, 0)$(1, 2, 0)
$(1, 3, 0)$(1, 3, 0)
$(\pi, e, 0)$(π, e, 0)

1 point

Consider three vectors $v_1 = (1, 0, 0), v_2 = (1, 1, 0), v_3 = (1, 1, 1)$v1 = (1, 0, 0), v2 = (1, 1, 0), v3 = (1, 1, 1) in the vector space R^3 R3, with usual addition and scalar multiplication. Which of the following option is true?

[Hint:Hint: Find out the vectors of the given sets in the options and form the matrices using the vectors as the columns and calculate the determinant of the matrices.]

○
$\{v_1, v_2, v_3\}${v1, v2, v3} is a linearly dependent set.
○
$\{v_1 + v_2, v_1 + v_3, v_3\}${v1 + v2, v1 + v3, v3} is a linearly dependent set.
○
$\{v_2 - v_1, v_1, v_2\}${v2 − v1, v1, v2} is a linearly dependent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{v_2 - v_1, v_1, v_2\}${v2 − v1, v1, v2} is a linearly dependent set.

Key points: 3Key points: 3

- Any set of nn vectors in R^2 R2 with $n \geq 3$n ≥ 3 is linearly dependent.
- Any set of nn vectors in R^3 R3 with $n \geq 4$n ≥ 4 is linearly dependent.
- Any set of $n + 1$n + 1 vectors in R^n Rn is linearly dependent.
- Any set containing a linearly dependent set is linearly dependent.
- If a set is linearly independent, then every subset of it, is linearly independent.

1 point

The subset $S = \{(3, 1), (1, 2), (-3, 1)\}$S = {(3, 1), (1, 2), (−3, 1)} of R^2 R2 is

[Hint: Hint: Any set of nn vectors in R^2 R2 with $n \geq 3$n ≥ 3 is linearly dependent.]

○ linearly dependent.
○ linearly independent.
○ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
linearly dependent.

1 point

Set $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$S = {(1, 1, −1), (−1, 1, 1), (0, 21, 0), (0, 1, −2), (1, 0, −2)} subset of R^3 R3 is

[Hint:Hint: Any set of nn vectors in R^3 R3 with $n \geq 4$n ≥ 4 are linearly dependent.]

- ○ linearly dependent.
- ○ linearly independent.
- ○ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

1 point

Let SS be a linearly dependent subset of R^3 R3. Which of the following options are true?

- ☐ $S \cup \{(1,0,0)\}$S ∪ {(1, 0, 0)} must be linearly dependent.
- ☐ $S \cup \{v\}$S ∪ {v} must be linearly dependent for any $v \in R^3$ v ∈ R3.
- ☐ $S \setminus \{v\}$S \ {v} must be linearly dependent for any $v \in S$v ∈ S.
- ☐ There may exist some $v \in S$v ∈ S, such that $S \setminus \{v\}$S \ {v} is still linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \cup \{(1,0,0)\}$S ∪ {(1, 0, 0)} must be linearly dependent.

$S \cup \{v\}$S ∪ {v} must be linearly dependent for any $v \in R^3$ v ∈ R3.

There may exist some $v \in S$v ∈ S, such that $S \setminus \{v\}$S \ {v} is still linearly dependent.

Check Answers

Your score is: 0/12